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Antibiotic prescribing trends among Belgian dentists (2014–2023) and the impact of the COVID-19 pandemic

Moira Kelly^{1*}, Roos Leroy², Thomas Struyf³, Boudewijn Catry^{1,4} and Lucy Catteau^{1,5}

Abstract

Background In 2016, dentists accounted for 5.8% of all outpatient antibiotic prescriptions in Belgium. This study evaluated antibiotic prescribing trends in dental practice from 2014 to 2023 in Belgium and explored the influence of prescriber demographics (age, sex and specialty), and of the COVID-19 pandemic on prescribing behaviour.

Methods Data on reimbursed systemic antibiotics (ATC J01) prescribed by dentists and dispensed by Belgian community pharmacies from 2014 to 2023 were quantified in Defined Daily Doses (DDDs) and expressed as DDD per 1000 insured beneficiaries per day (DBD). This was combined with data on dentist demographics (age, sex, specialty) and professional activity (annual number of consultations for 2018–2022) to assess prescribing behaviour.

Results Total antibiotic prescriptions by Belgian dentists (measured in DDD) increased by 13.5% from 2014 to 2023. Despite a 3.1% reduction in DDD in 2020, the volume of J01 antibiotics prescribed per consultation increased during that year. The proportion of total outpatient J01 antibiotic deliveries prescribed by dentists increased from 4.9% in 2014 to 6.4% in 2023, with a peak of 7.3% in 2021. Male and older dentists consistently prescribed more antibiotics than their younger and female counterparts. Following the introduction of national antibiotic guidelines for dentistry in 2020 we observed limited change in the proportion of recommended antibiotics prescribed.

Conclusion Antibiotic prescribing among Belgian dentists is influenced by various factors, including prescriber characteristics and external disruptions such as the COVID-19 pandemic. Opportunities remain to optimize prescribing practices, and efforts should focus on encouraging guideline-based antibiotic use only when necessary.

*Correspondence:

Moira Kelly
moira.kelly@sciensano.be

¹Department of Epidemiology and Public health, Sciensano, Brussels, Belgium

²Belgian Health Care Knowledge Centre (KCE), Brussels, Belgium

³Academic Centre for Family Medicine, KU Leuven, Leuven, Belgium

⁴Faculty of Medicine, Université Libre de Bruxelles, Brussels, Belgium

⁵Faculty of Medicine and Pharmacy, Université de Mons (UMons), Mons, Belgium



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Text box 1. Contributions to literature

- This study confirms that dentists are major contributors to outpatient antibiotic use in Belgium. Their share increased from 2014 to 2023, and peaked during the COVID-19 pandemic, probably due to greater decreases in antibiotic prescriptions in other outpatient settings.
- Despite national efforts such as treatment guidelines and consumption targets, no improvements in prescribing volumes or antibiotic choice were observed within the available timeframe.
- Identifies male and older dentists as prescribing more antibiotics than their female and younger colleagues.
- Proposes “Defined Daily Dose per consultation” as a more accurate metric to assess prescribing behaviour.
- These findings point to the need for more targeted, data-driven stewardship strategies for dental professionals.

Introduction

Antimicrobial resistance (AMR) poses a critical global health challenge, exacerbated by the overuse and misuse of antibiotics across healthcare settings. Although Belgium has witnessed a significant reduction in overall consumption of antimicrobials for systemic use over the past decade, the country still faces high usage rates compared to the European population mean [1]. Notably, while Belgian hospitals demonstrate consistently favourable usage rates, the ambulatory care sector remains a significant contributor to antibiotic consumption, accounting for 82% of total usage in 2023 [1, 2].

Within the Belgian ambulatory care landscape, dentists have emerged as notable contributors to antibiotic prescribing practices. In 2016, they prescribed 5.8% of total J01 antibiotics consumed, with significant contributions for specific antibiotics such as clindamycin (20.1%), metronidazole (11.6%), amoxicillin (10.5%), and amoxicillin combined with enzyme inhibitor (8.4%) [3]. Inappropriate prescribing of antibiotics in dentistry is reported from several parts of the world underscoring the importance of addressing antibiotic prescribing behaviours within the dental community [4–7].

In response to these concerning trends, Belgium launched a national action plan in 2020 to combat AMR, with targeted efforts aimed at reducing antimicrobial consumption and promoting responsible prescribing practices among healthcare providers [8]. In the same year, the Belgian Health Care Knowledge Centre (KCE) published an evidence-based clinical guideline to help dentists make informed decisions about the prudent prescription of systemic antibiotics in 12 clinical indications (including irreversible pulpitis, replantation of avulsed permanent teeth, dental implant placement, and antibiotic prophylaxis in patients at (high) risk of infective endocarditis undergoing dental procedures) [9]. This guideline emphasises that dental treatment is in most of the studied indications the most effective approach to manage infections, with antibiotics rarely necessary.

When antibiotics are needed, the guideline recommends the use of narrower-spectrum antibiotics such as amoxicillin [9]. Across all systematically assessed clinical scenarios, only six antibiotics are recommended (i.e. amoxicillin (or ampicillin), cefuroxime axetil, azithromycin, clarithromycin, metronidazole, and clindamycin). Two years after the publication of the guideline, the National Institute for Health and Disability Insurance (NIDHI) provided all dentists with personalized feedback on their antibiotic prescribing behaviour to raise awareness and compare practices [10].

In 2020, the COVID-19 pandemic led to widespread lockdowns and restrictions, significantly impacting healthcare delivery worldwide, including dentistry. The Federal and regional governments, together with federal scientific and medical bodies recommended postponing all non-urgent healthcare appointments to reduce the risk of COVID-19 transmission and prioritise resources towards providing optimal care for patients with COVID-19 and urgent conditions [11]. Dentists adapted to this situation by implementing tele-dentistry and pharmacotherapy to triage and manage dental emergencies during the pandemic's early stages [12]. Such restrictions may have influenced dentists' antibiotic prescribing behaviour, with varying effects observed globally [11–13]. Impacts continued outside of lockdowns, with mandatory waiting and ventilation periods restricting the number of patients.

This study aims to evaluate the antibiotic prescribing trends of Belgian dentists from 2014 to 2023, including particular focus on the COVID-19 years of 2020 and 2021. Additionally, it seeks to explore variations in prescribing practices across different antibiotic agents and dentist specialties. For the first time in Belgium, the Defined Daily Dose per consultation (DDD/consult) metric was used to analyse dental prescribing trends, offering a new perspective on antimicrobial use relative to clinical activity. By shedding light on these dynamics, the study aims to provide crucial information for refining antibiotic stewardship efforts within the Belgian dental community and addressing the broader challenge of AMR.

Materials and methods**Study design**

This retrospective observational study analyzed antimicrobial prescribing trends among Belgian dentists from 2014 to 2023. Data were sourced from the Pharmanet database managed by the National Institute for Health and Disability Insurance (NIDHI) and from the Intermunicipal Agency (IMA) [13, 14]. The data were already anonymized, de-identified and aggregated by the data provider (NIDHI) or publicly available (IMA), ensuring compliance with privacy regulations, as such ethical review was not needed.

Data sources

Several data sources were used to provide a comprehensive view of antibiotic prescribing practices:

Prescription data (2014–2023)

Collected from the Pharmanet database, these data comprised reimbursement information from the seven health insurance funds in Belgium, covering 99% of the Belgian population in 2023 [13]. The dataset included dental prescriptions that were dispensed and collected by patients at community pharmacies each year from 2014 to 2023. Prescriptions written by maxillofacial surgeons and those delivered by hospital pharmacies for hospitalized patients were excluded from this study. In addition to these dental prescriptions, all reimbursed prescriptions from the community sector were included to allow for comparative analyses.

Antimicrobials were classified according to the Anatomical Therapeutic Chemical (ATC) classification system of the World Health Organization (WHO) with volumes expressed in the number of packages prescribed and Defined Daily Dose (DDD). The number of packages prescribed refers to the number of reimbursed boxes of antibiotics delivered by pharmacies – as antibiotics in Belgium are generally prescribed as a full box, with different sized boxes available per molecule [15]. DDD was calculated using the 2023 DDD/ATC Index [16]. The analysis focused on all medications within the ATC J01 group (antibacterials for systemic use), as well as metronidazole (P01AB01). The dataset excluded all non-reimbursed medications, as well as magistral preparations.

General population data (2014–2023)

This dataset, obtained from the Intermutualistic Agency (IMA, www.aim-ima.be), comprises data on the number of insured beneficiaries per province, covering approximately 99% of the Belgian population. The total number of insured beneficiaries in Belgium was used to control for changes in population size when considering the volume of antimicrobials consumed e.g. for calculating DDD per 1000 insured beneficiaries per day (DBD).

Detailed prescriber datasets, described below, were only available for the 5-year period 2018–2022

Prescriber data (2018–2022)

Provided by NIDHI, detailing the number of active prescribers each year from 2018 to 2022. Dentists were categorized by age (in five-year intervals: 35 years and under, 36–40, 41–45, 46–50, 51–55, 56–60, and over 61 years), sex and specialty (general dentists, periodontists, and orthodontists). Active prescribers were defined as those with at least one patient assigned to their case load by NIHDI, meaning that for at least one patient they were recorded as a patient's "main" dentist for that specialty.

Maxillofacial surgeons were excluded from the present analyses as in Belgium they rarely perform "regular dental/periodontal/orthodontic care". Dentists in training were also excluded. To avoid small cells, any record with fewer than 5 practitioners in a year was noted as "<5", and these records were replaced with 3 to facilitate their inclusion in analyses. Such cells made up 8 of 272 data points, with small cells representing 1.0–2.0% of periodontists each year.

Consultation data (2018–2022)

Also provided by NIDHI, this dataset includes the number of patient contacts performed by dentists each year, from 2018 to 2022, aggregated by prescriber's age (in five-year intervals: 35 years and under, 36–40, 41–45, 46–50, 51–55, 56–60, and over 61 years), sex and specialty. All consultation types were included in this dataset – calculated as the number of distinct days on which any nomenclature code was reimbursed (including all consultation types, physical and tele-consultations). Note that, as calculated by NIHDI, each patient is assigned to a "main" practitioner within each specialty. This designation represents the dentist seen most frequently by the patient in that year, within that specialty. All consultations for that patient within that specialty are then assigned to this "main" dentist for the corresponding year.

Antimicrobial consumption metrics

The primary metric for antimicrobial consumption was quantified as Defined Daily Doses (DDD) and DDD per 1000 beneficiaries per day (DBD). In addition, the number of packages prescribed was considered – combining this with DDD to consider the average DDD per package. In addition to total consumption of selected ATC-codes relevant to dentistry: J01 and P01AB, several secondary indicators were calculated to assess prescribing quality, following national and international guidelines encouraging fewer broader-spectrum antimicrobial treatments and preserving newer generation antimicrobials [1, 8]. These include:

1. the ratio of amoxicillin to amoxicillin-clavulanic acid consumption: this indicator encourages higher proportional prescription of amoxicillin, a narrower-spectrum antibiotic, over amoxicillin-clavulanic acid, broader-spectrum.
2. The percentage of antimicrobial deliveries by total volume (DDD) comprising antibiotics included in the national treatment guidelines published by the Belgian Health Care Knowledge Centre (KCE) [9] – namely amoxicillin (J01CA04), ampicillin (J01CA01), azithromycin (J01FA10), clarithromycin (J01FA09), metronidazole (P01AB01, J01XD01), cefuroxime (J01DC02), clindamycin (J01FF01). We assessed

the proportion of dental antibiotic prescribing that aligned with these recommendations before and after the guideline was released.

Statistical analyses

All statistical analyses were performed using R version 4.2.1. Data cleaning and visualisation were performed using the *tidyverse* library [17]. A linear mixed-effect model was used to analyse DDD per consultation of total J01 medications among general dentists, considering prescriber age (in 10 year age groups- assigned as an ordered categorical variable) and sex, with year as a random effect, using the *lme4* package and maximum likelihood (restricted maximum likelihood (REML) = FALSE). The best fitting model had formula $DDD_per_consult \sim Age*sex + (1|Year)$. Model comparisons were performed using the *anova()* function, and the effect of random variables was estimated using parametric bootstrapping. Models were visualised using the *sjPlot* package functions *plot_model()* and *tab_model()*, and model residuals were visualised using functions from the *stats()* package. Pearson's product-moment correlation was calculated using the *cor()* and *cor.test()* functions, with assumptions of normality first tested using the *shapiro.test()* function and *qqplots*.

Results

Antimicrobial prescriptions in dental practice, 2014–2023

Over the past decade, the total volume of J01 antibiotics prescribed by Belgian dentists increased by 13.6% in DDDs (Table 1), and by 8.7% in DBD, with some year-to-year fluctuations. In contrast, the number of antibiotic packages of antimicrobials prescribed declined by -5.4% over the same period. This inverse trend suggests a moderate shift towards prescribing larger package sizes, increasing by approximately one DDD per package on average from 2013 to 2024. A greater increase was observed in the prescription of 'Antiprotozoals' (P01AB, +25.1% DDD, +19.6% DBD) across the study period, with a dip in 2020. Metronidazole accounted for 99.7% of all prescribed antiprotozoals.

Among J01 antibacterials, amoxicillin and amoxicillin with an enzyme inhibitor were the most commonly prescribed, accounting for 49.3% and 39.0% respectively of all J01 antibacterials prescribed by Belgian dentists in 2023. The DBD for amoxicillin (J01CA04) increased by 16.7% between 2014 and 2023, with a notable decrease of 4.1% between 2019 and 2020, followed by a 7.6% increase from 2020 to 2021, surpassing pre-pandemic levels. In contrast, the DBD for amoxicillin with enzyme inhibitor (J01CR02) rose by 21.1% from 2014 to 2023, without any decline noted in 2020. As a result, the amoxicillin to amoxicillin-clavulanic acid ratio [DBD(J01CA04)/DBD(J01CR02)] decreased from 1.31 in 2014 to 1.26 in

2023 (Pearson's product-moment correlation $cor = -0.64$, $p = 0.048$) with a notable trough during the COVID-19 pandemic in 2020 (1.15), suggesting a relative increase in the use of combination therapy.

Following the publication of the national guidelines in 2020, the proportion of dental antimicrobial prescriptions corresponding to antibacterials recommended within the national guidelines remained largely unchanged. The largest proportion of prescribed antimicrobials not included in the guidelines corresponds to amoxicillin with enzyme inhibitor (J01CR02), representing 39% of dental J01 antibiotic prescriptions in 2023. Three other antibiotics (doxycycline, nitrofurantoin and spiramycin) are listed in the top 10 most prescribed molecules by dentists in 2023, yet they are not included in the national guideline. Notably, the DBD for clarithromycin (recommended in national guidelines) declined by 49.9% from 0.044 in 2014 to 0.022 in 2023, indicating a more cautious approach toward its use. The DBD rate for penicillin V remained very low (0.0001; not presented in Table 1).

Between 2014 and 2023, dental prescriptions accounted for an increasing share of the total J01 antibiotic use in ambulatory care, rising from 4.9% to 6.4%, with a peak of 7.2% and 7.3% in 2020 and 2021 respectively. The relative contribution of dental prescription was particularly high for amoxicillin and amoxicillin with an enzyme inhibitor (on average 12.1% and 9.5% of total Belgian ambulatory use), increasing across the study period, with notable peaks in 2020 (16.2% for amoxicillin and 12.4% amoxicillin with an enzyme inhibitor). High relative consumption was also noted for clindamycin (16.9% in 2023), spiramycin (24.1%) and metronidazole (11.5%). In contrast, the relative contribution of penicillin V was very low (0.0000315%).

Trends in the dental workforce and dentist activity (2018–2022)

Our population data portray a small increase in the number of active dentists in Belgium (+1.1% from 2018 to 2022), while in the same period the proportion of female dentists rose from 51.3% to 53.9% (Table 2). Male dentists are only in the majority in the +55 age groups among the general dentists and periodontists (Fig. 1). From 2018 to 2022, a slight increase in the number of insured inhabitants was also noted (+2%). The number of dental consultations also increased during the study period, reflecting the rising number of active dentists and the relatively stable frequency of dental visits per insured inhabitant. However, in 2020, a substantial decline of approximately 15.8% in the number of dental consultations per inhabitant occurred, which recovered to pre-pandemic levels by 2022.

Table 1 Antimicrobials prescribed by Belgian dentists (including general dentists, periodontists and orthodontists) in the ambulatory care setting, and the proportion (%) dentistry-related prescriptions contribute to total ambulatory use, 2014–2023

Antimicrobial Group / Product	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	Percent change 2014–2023 (%)	Percentage of all J01 dental prescriptions in 2023 ^a
DDD (thousand)												
Antibacterials for systemic use (J01)	4580.2	4678.9	4700.5	4605.8	4569.1	4710	4563	4842.1	4995.2	5202.7	13.59	/
Antiprotozoals (P01AB)	55	59.2	59.4	60	61.2	66.1	55.1	65	66	68.8	25.09	/
Packages (thousand)	662	667 ^b	666 ^b	638 ^b	620 ^b	620 ^b	579 ^b	603 ^b	609 ^b	626 ^b	-5.43	/
Antibacterials for systemic use (J01)												
DBD (Proportion of total ambulatory consumption prescribed by dentists %) ^c												
Total Antibacterials for systemic use (J01)	1.139 (4.9%)	1.158 (4.9%)	1.159 (5.1%)	1.131 (5.3%)	1.118 (5.4%)	1.147 (5.8%)	1.107 (7.2%)	1.171 (7.3%)	1.196 (6.3%)	1.238 (6.4%)	8.69	100
Amoxicillin (J01CA04)	0.522 (10.8%)	0.536 (10.7%)	0.537 (10.6%)	0.525 (11.0%)	0.519 (10.8%)	0.533 (11.0%)	0.511 (16.2%)	0.549 (15.7%)	0.574 (11.9%)	0.61 (12.1%)	16.73	49.3
Amoxicillin and enzyme inhibitor (J01CR02)	0.398 (7.1%)	0.412 (7.2%)	0.422 (8.5%)	0.419 (8.8%)	0.425 (8.4%)	0.445 (9.4%)	0.446 (1.2%)	0.467 (1.2%)	0.474 (10%)	0.482 (11%)	21.21	39.0
Clindamycin (J01FF01)	0.078 (22.5%)	0.077 (21.5%)	0.076 (20.3%)	0.073 (19.3%)	0.072 (17.9%)	0.072 (17.3%)	0.068 (18.1%)	0.071 (19.3%)	0.069 (17.8%)	0.068 (16.9%)	-12.78	5.4
Azithromycin (J01FA10)	0.025 (1.7%)	0.026 (1.6%)	0.027 (1.5%)	0.027 (1.5%)	0.026 (1.3%)	0.027 (1.3%)	0.027 (1.7%)	0.028 (1.7%)	0.029 (1.3%)	0.030 (1.3%)	23.05	2.4
Clarithromycin (J01FA09)	0.044 (2.9%)	0.040 (2.7%)	0.040 (2.8%)	0.036 (3.0%)	0.033 (2.7%)	0.030 (2.7%)	0.025 (3.6%)	0.025 (4%)	0.023 (2.8%)	0.022 (2.5%)	-49.85	1.8
Metronidazole (anti-Protozoals) (P01AB01)	0.014 (11.2%)	0.015 (11.7%)	0.015 (11.6%)	0.015 (11.5%)	0.015 (11.8%)	0.016 (12.3%)	0.013 (11.7%)	0.016 (12.4%)	0.016 (11.7%)	0.016 (11.5%)	20.24	/
Doxycycline (J01AA02)	0.037 (4.9%)	0.034 (4.5%)	0.032 (4.2%)	0.029 (4.0%)	0.025 (3.4%)	0.023 (3.1%)	0.018 (3.3%)	0.016 (2.4%)	0.014 (1.9%)	0.013 (1.6%)	-65.20	1.0
Cefuroxime (J01DC02)	0.006 (0.4%)	0.005 (0.4%)	0.005 (0.4%)	0.005 (0.5%)	0.005 (0.4%)	0.005 (0.4%)	0.004 (0.6%)	0.004 (0.6%)	0.004 (0.5%)	0.004 (0.5%)	-23.55	0.4
Nitrofurantoin (J01XE01)	0.003 (0.2%)	0.003 (0.2%)	0.003 (0.2%)	0.003 (0.2%)	0.003 (0.1%)	0.004 (0.2%)	0.003 (0.1%)	0.003 (0.2%)	0.003 (0.1%)	0.003 (0.2%)	18.94	0.3
Spiramycin (J01FA02)	0.006 (29.6%)	0.006 (28.4%)	0.005 (27.7%)	0.005 (29.0%)	0.004 (27.3%)	0.004 (28.5%)	0.003 (23.2%)	0.002 (24.2%)	0.002 (24.9%)	0.002 (24.1%)	-67.06	0.2
Secondary Indicators – calculated based on DBD												
Amoxicillin:amoxicillin-clavulanic acid	1.31	1.30	1.27	1.25	1.22	1.20	1.15	1.18	1.21	1.26	-3.82	/
Proportion antibiotics listed in national guide-lines (%) [*]	59.61	59.56	59.59	59.28	59.05	58.68	57.76	58.44	58.97	59.81	0.34	/

DDD Defined daily doses, DBD Defined Daily Doses per 1,000 Beneficiaries per Day

^aPercentage of all reimbursed J01 antibiotics prescribed by dentists in 2023, calculated using DBD^bExcludes antibiotic prescriptions to nursing home residents delivered as “units” rather than packages – representing <0.1% of antibiotic deliveries (in DDD) annually^cUpper line = DBD values for each year, lower line in parentheses = the percentage of ambulatory-sector reimbursed antibiotic consumption prescribed by dentists for the relevant metric. For “Total Antibacterials for systemic use (J01)” this indicates the proportion of all ambulatory J01 reimbursed antibiotics that were prescribed by dentists, for the specific antibiotics this represents the proportion of the total volume of that antibiotic reimbursed within the ambulatory sector that was prescribed by dentists, calculated using DBD. For these rows “Percent change 2014–2023 (%)” refers to the change in DBD^{*}Percentage of total annual J01 and P01AB prescribed by dentists that are included in the national treatment recommendations for dentists, calculated using DBD

Table 2 Overview of the number of active dentists, insured inhabitants, and dental consultations in Belgium, (2018–2022)

Year	Number of active dentists* (Male%)	Number of insured inhabitants [†] (million)	Number of dental consults* (million)	Dentists per 1,000 insured inhabitants	Dental consults per inhabitant	Dental consults per dentist
2018	8449 (48.7%)	11.19	13.28	0.76	1.19	1572
2019	8610 (48.3%)	11.25	13.47	0.77	1.20	1564
2020	8663 (47.5%)	11.29	11.36	0.77	1.01	1311
2021	8545 (46.8%)	11.33	13.29	0.76	1.17	1555
2022	8545 (46.1%)	11.44	13.66	0.75	1.19	1599

Sources: *National Institute for Health and Disability Insurance (NIDHI) – active dentist identified by being the assigned dentist to at least one patient, including general dentists, periodontists and orthodontists, see methods for more details

[†] InterMutualistic Agency (IMA)

Antimicrobial prescriptions considering prescriber population and activity

When considering prescribing behaviour in the context of the number of prescribers and the number of dental contacts with patients, we observe an overall increase in the DDD per prescriber for J01 antibiotics from 540.8 in 2018 to 584.6 in 2022 (+8.1%), though a drop to 526.7 DDD per prescriber was observed in 2020 (-3.7% compared to 2019, Table 3). By contrast, the DDD per consultation showed a significant peak in 2020, increasing by 14.8% from 2019 to 2020.

Further analysis reveals notable variations based on prescriber characteristics (Fig. 2). Among general dentists, age (χ^2 [1] = 43.0, $p = 1.2e-7$) and sex (χ^2 [1] = 115.6, $p = 2.2e-16$) were significantly associated with the volume of antibiotics prescribed per consult (general mixed effect model where DDD per consult was explained by age (ordered categorical variable) and sex, considering year as a random effect, see Supplementary Fig. 1). Overall, periodontists exhibited the highest DDD/consult rates, steadily increasing from 0.67 DDD/consult in 2018 to 0.84 DDD/consult in 2022 (see Supplementary Table 1). General dentists maintained relatively stable prescribing rates around 0.4 DDD/consult, with a slight peak in 2020. In contrast, orthodontists consistently had near zero antimicrobial prescribing rates. Although periodontists prescribed more antimicrobials per consult, they prescribed significantly higher proportions of recommended antimicrobials included in national guidelines compared to general dentists (Supplementary Table 2, mean percentage of recommended antimicrobials: 75.5% vs. 58.4%, paired student t-test $t = 53.0$, $p = 7.5 \times 10^{-13}$).

When considering all dentists, the patterns most closely resemble the behaviour of general dentists as they make up over 91% of the dentist population and 95% of the dental antimicrobial prescriptions each year. Dentists in the oldest age group (≥ 61 years) consistently had the highest prescribing rates, peaking at 0.48 DDD/consult in 2020, while younger dentists (< 35 years) peaked at 0.34 DDD/consult that same year. Male dentists consistently prescribed more antibiotics than female dentists, with a

17% increase in DDD/consult between 2019 and 2020, reaching 0.48 DDD/consult in 2020. In contrast, female dentists saw a 13% increase in DDD/consult during the same period, peaking at 0.32 DDD/consult in 2020, and returned to pre-pandemic levels by 2021, whereas male prescribing rates remained elevated at 0.45 DDD/consult in 2021/2022.

Discussion

This study analysing antimicrobial reimbursement data found that antibiotics prescribed by Belgian dentists accounted for 6.4% of total systemic antibiotic consumption in ambulatory care in 2023, an increase compared to the last published value of 5.8% in 2016 [3]. The overall volume of antibiotics prescribed by Belgian dentists expressed in DBD, has also increased from 2014 to 2023. While dentists account for a relatively small proportion of total ambulatory J01 antibiotic consumption (6.4% in 2023), they represent the second largest prescribing profession in Belgian outpatient care [18]. Furthermore, dental prescribing accounts for a disproportionately high share of certain antibiotics, including amoxicillin-clavulanic acid, for which dentists prescribed 11% of total ambulatory consumption in 2023, an antibiotic that is central to national targets aimed at improving prescribing quality [8].

These results highlight limited progress in optimizing prudent prescribing of antibiotics and antibiotic selection, despite efforts to reduce antimicrobial use, including the publication of a National Action Plan against antimicrobial resistance (NAP-AMR, although it remains unclear to what extent this document was communicated to front line stakeholders such as dentists) [8] and national guidelines for antibiotic use in dental practice [9]. Notably, the NAP-AMR includes targets to reduce the absolute and relative consumption of broad spectrum antibiotics and amoxicillin-clavulanic acid specifically. Similarly, national guidelines explicitly exclude amoxicillin-clavulanic acid for the 12 common dental indications under study, yet it remains widely prescribed. Among antibiotics prescribed, amoxicillin and

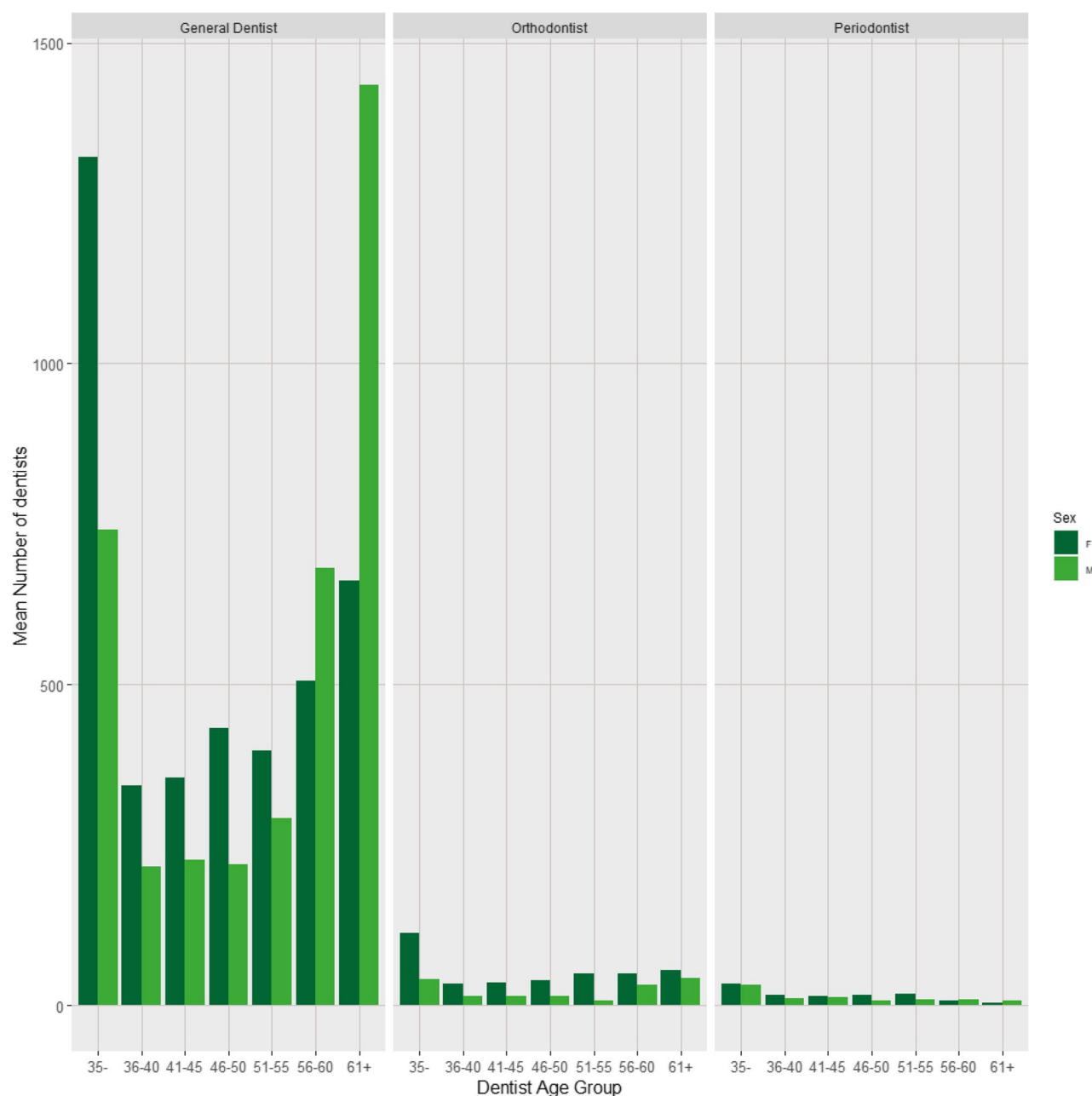


Fig. 1 Mean annual number of active dentists stratified by age group, sex and specialty, 2018–2022, Belgium. F = female, M = male

Table 3 Antimicrobials (ATC J01 and P01) prescribed by Belgian dentists in the ambulatory setting expressed as DDD per prescriber and DDD per consultation (2018–2022)

	2018	2019	2020	2021	2022
DDD per prescriber					
Antibacterials for systemic use (J01)	540.8	547.0	526.7	566.7	584.6
Antiprotozoals (P01AB)	7.25	7.68	6.36	7.61	7.73
DDD per consultation					
Antibacterials for systemic use (J01)	0.34	0.35	0.40	0.36	0.37
Antiprotozoals (P01AB)	0.005	0.005	0.005	0.005	0.005

DDD Defined daily dose, ATC Anatomical Therapeutic Chemical, with DDD calculated according to the WHO ATC/DDD index (https://atcddd.fhi.no/atc_ddd_index/)

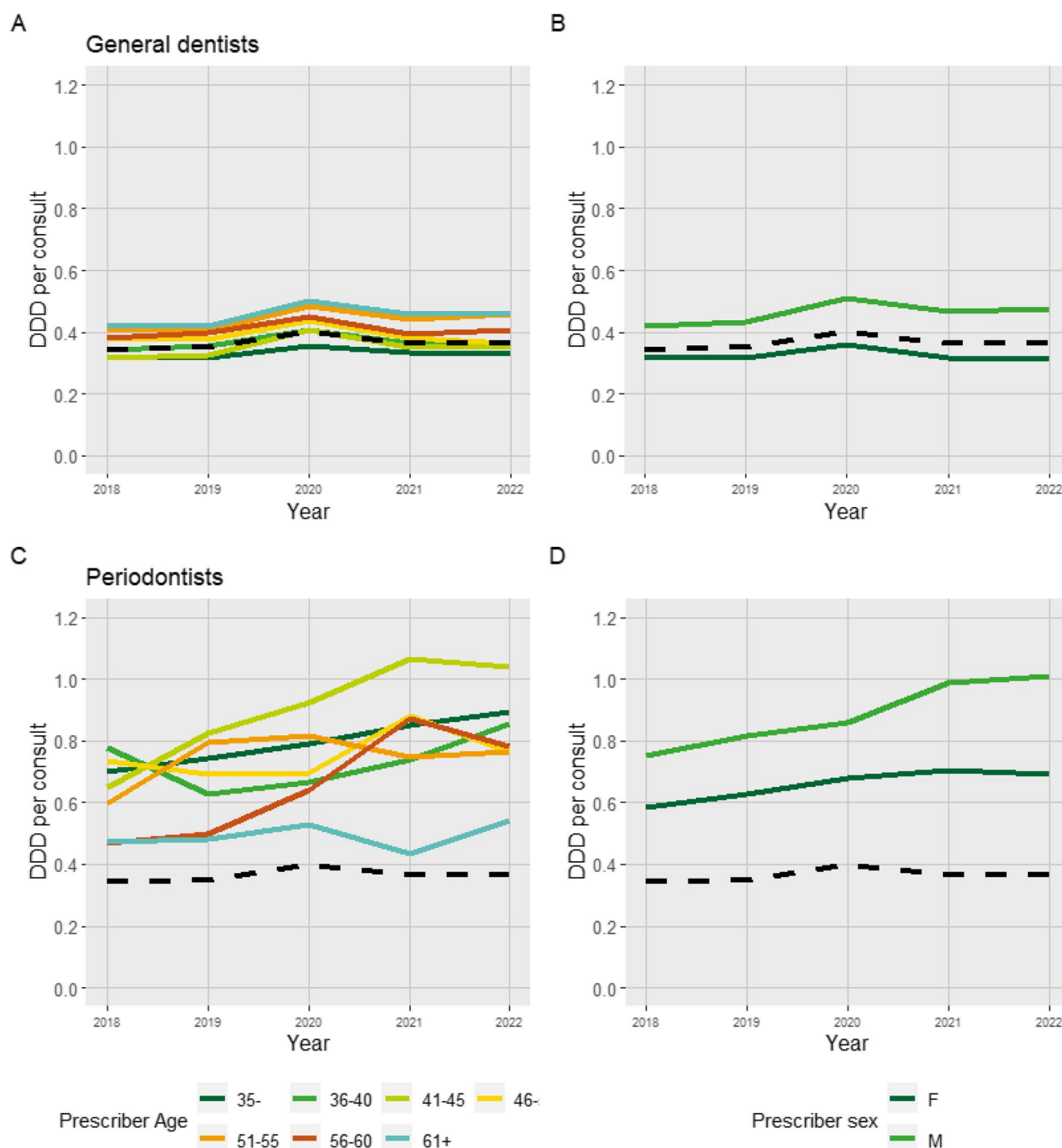


Fig. 2 Volume of J01 antibiotics, expressed as Defined Daily Doses (DDD) per consultation, prescribed by Belgian dentists, stratified by age group for general dentists (**A**) and periodontists (**C**), and sex for general dentists (**B**) and periodontists (**D**) from 2018 to 2022. DDD calculated using the 2023 DDD/ATC Index [14], compared to the average across all dentists (black dashed line). Orthodontists are not shown as they prescribe less than 1% of antimicrobial prescriptions each year. ATC: Anatomical Therapeutic Chemical classification system, DDD calculated using WHO ATC/DDD index (https://atcddd.fhi.no/atc_ddd_index/); F: female; M: male

amoxicillin-clavulanic acid are the most commonly used, representing 88.3% of total reimbursed dental J01 antibiotic deliveries in DDD in 2023. This pattern has remained stable compared to previous analyses [3], with no substantial shift towards narrower spectrum antibiotics and

no improvement in the ratio of amoxicillin / amoxicillin-clavulanic acid (1.27 in 2016, 1.26 in 2023) which remains far from the national target of 4.

Furthermore, while the Belgian Health Care Knowledge Centre (KCE) recommended in 2019 to move to a

practice of delivering the exact number of antibiotics for the recommended treatment course [19], there has been no movement in this direction and conversely we observe an increase in the DDD per package. Availability of smaller sized packages may impact this measurement, and although moderate (one additional DDD per package from 2014 to 2023), it is concerning to note this pattern of increasing DDD per package has been observed in Belgium for over 20 years [3, 20].

Such challenges in improving dental prescribing behaviours appear common to many countries, with national-level studies often observing high, and increasing dental prescription of amoxicillin-clavulanic acid [21, 22] with dental antibiotic prescribing patterns not responding to reduction initiatives [23]. It is too early to assess the impact of more recent efforts such as the provision in 2022 of individual feedback on dentist antimicrobial prescription [10].

One complicating factor in evaluating the impact of efforts to reduce antibiotic consumption is the COVID-19 pandemic, which coincided with the launch of both the NAP-AMR and the publication of the national guidelines in 2020. In ambulatory care (medical and dental), there was a notable decrease in antibiotic consumption of 22.9% from 2019 (19.98 DDD per 1000 inhabitants per day (DID)) to 2020 (15.45 DID) [24]. This decline was observed in other countries too and ascribed to reduced antimicrobial prescriptions for respiratory tract infections [25]. COVID-19 public health interventions (e.g. increased social distancing, face mask-wearing and hand hygiene) interrupted seasonal peaks in communicable respiratory pathogens [26]. These interventions have no impact on oral health, and we observe a 24.1% increase in the proportion of ambulatory care antibiotics attributed to dental prescriptions from 2019 to 2020 (5.8% in 2019, 7.2% in 2020). Furthermore, the proportion of broader spectrum amoxicillin-clavulanic acid prescriptions increased, and while individual dentists were prescribing less overall (-3.7% in DDD/prescriber from 2019 to 2020), possibly due to lockdowns and fewer patients, the patients they did see were more likely to receive an antibiotic (+ 13.1% in DDD/consult from 2019 to 2020), possibly due to restricted physical procedures or more severe presentations at the time of contact. Similar increases were observed in other European nations such as the UK and Norway [21, 27]. Several factors may have contributed to these trends, including (i) the postponement of non-urgent dental procedures, leading to a higher proportion of consultations involving acute infections; (ii) patient reluctance to seek care, resulting in more severe cases upon presentation, and (iii) concerns about viral transmission, which may have led dentists to favour pharmacological interventions over aerosol-generating procedures [28]. While other studies observed the greatest

increase in dental antibiotic prescriptions during the period with most restricted access to care [29], we were not able to assess this in Belgium with the data available.

Beyond temporal variations, prescriber demographics also influence antibiotic prescribing behaviour. Male dentists and older dentists (particularly older, female dentists) prescribe higher volumes of antibiotics, particularly when considering DDD per consultation. This pattern aligns with prior studies in primary care, which report higher rates of antibiotic prescribing [30] and longer treatment durations [31] among older general practitioners, and reports that practitioners under the age of 40 were more likely to follow clinical guidelines than those over the age of 60 [32]. It may also be that female dentists are more hesitant to prescribe antibiotics - a previous study found that female general practitioners were less likely to prescribe antibiotics in the first instance than their male counterparts when treating the same clinical presentation (sore throat symptoms) [33], more often adopting a “wait-and-see” approach. The patient profiles of younger and female Belgian dentists may also vary compared to older and male dentists, and patient profile can also play a significant role in the likelihood of receiving antibiotic treatment [30, 33]. Whether our observations are (in part) due to a variability in patient profiles and/or due to different prescribing behaviour cannot be deduced from our data, but could be explored in further research.

Compared to previous analyses, a key methodological advancement of this analysis is the inclusion of consultation data, allowing for more refined assessment of prescribing behaviour that accounts for activity level. For example - while dentists under 35 years of age prescribed the second largest volume of antibiotics (after those aged over 61 years) in 2022 when expressed in DBD, they had the lowest prescribing rate when expressed as DDD per consultation - indicating a higher activity level and proportionally lower antibiotic use per consultation.

Nevertheless, this study has several limitations. Firstly, by definition, reimbursement data do not include non-reimbursed antibiotics, including magistral preparations and imported products. We cannot directly estimate the volume of non-reimbursed antibiotic use; however, its impact on our results is likely to be limited. The most important potential contributor is fluoroquinolones - for which reimbursement restrictions were introduced in 2018 [34]. Yet before these restrictions, fluoroquinolones represented on average only 0.44% of dental antibiotic prescriptions (annual average: 20,233 DDD), decreasing to 0.03% after the change in reimbursement criteria. Using national ambulatory sales data (irrespective of reimbursement status and prescribing attribution), we observed a 41% reduction in fluoroquinolone sales between 2017 and 2023. Assuming a similar proportional

reduction in dental prescribing, non-reimbursed fluoroquinolones would represent approximately 0.23% of antibiotics prescribed by dentists in 2023. This suggests that the exclusion of non-reimbursed antibiotics is unlikely to substantially influence our conclusions. Secondly, there are no data available on the indications for antibiotic prescriptions, currently making it impossible to assess the appropriateness of the treatment. The linkage of antibiotic prescription data to dental records or clinical indication in future studies would represent a significant advance in the ability to monitor and assess prescribing quality. Thirdly, a 2019 revision of the WHO- defined DDD for amoxicillin and amoxicillin- clavulanic acid (from 1 g to 1.5 g) [35] complicates direct comparisons between this study and previous analyses using this metric [3]. This does not impact analyses within our study as the 2019 definition is used throughout. Finally, the limited availability of the more detailed prescriber data (2018–2022) compared to the prescription data (2014–2023), prohibited our ability to consider longer-term trends and place the data from the COVID-19 years in a broader context.

Conclusion

The study demonstrates that antibiotic prescribing patterns among Belgian dentists are associated with a range of factors, including prescriber characteristics and external events such as the COVID-19 pandemic. Despite existing national guidelines, prescribing behaviour, particularly the continued use of broad-spectrum and non-recommended antibiotics, appears largely unaffected by recent efforts to reduce non-prudent antibiotic prescription. These findings underscore the need for targeted antimicrobial stewardship interventions that specifically address dental prescribing practices. Dentists should not be overlooked in national antimicrobial stewardship strategies. Integrating dentistry more fully into stewardship programs is essential to ensure a coordinated approach to combat antimicrobial resistance.

Abbreviations

AMR	Antimicrobial resistance
ATC	Anatomical Therapeutic Chemical classification system
DBD	Defined Daily Doses per 1000 insured beneficiaries per day
DDD	Defined Daily Dose(s). DDD is used to quantify antibiotic volumes
DDD/consult	Defined Daily Dose per consultation
DID	Defined Daily Doses per 1000 inhabitants per day
F	Female (used in figures)
IMA	Intermutualistic Agency
J01	ATC J01 group, which refers to antibacterials for systemic use
KCE	Belgian Health Care Knowledge Centre
M	male (used in figures)
NAP-AMR	National Action Plan against antimicrobial resistance
NH	Nursing Home
NIDHI	National Institute for Health and Disability Insurance
REML	Restricted maximum likelihood (used in statistical analysis)
WHO	World Health Organization

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

MK and LC conceptualised the manuscript, MK, LC and RL designed the methodology, MK performed the analysis and created the figures. MK and LC wrote the original manuscript and all authors read and revised the manuscript.

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Data availability

All data is available through contacting the National Institute for Health and Disability Insurance (prescription and prescriber dataset) or online at the Intermutuality Agency Atlas (<https://www.aim-ima.be/Atlas-AIM>)(<https://www.aim-ima.be/Atlas-AIM>), population dataset).

Declarations

Ethics approval and consent to participate

Ethical approval was not obtained for the study as all data included were aggregated at a national level, with small cell analyses ensuring data grouping prohibited patient or prescriber identification.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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